



AI DATA ANALYST AGENT

Professional Data Analysis Report

Generated On: 15 July 2026 11:40 PM

Developer: Pritiranjana Rout

Version: 2.0.0

EXECUTIVE SUMMARY

This report provides a comprehensive analysis of our workforce dataset, which contains demographic, departmental, and compensation metrics across 21 employee records. The primary objective is to evaluate overall data quality, uncover key operational patterns, and deliver strategic HR and business recommendations. Our findings indicate an exceptionally high Data Health Score of ****92****, though minor data-cleaning steps are required to address a single missing salary entry, one duplicate row, and one experience-related outlier. From a strategic standpoint, we observe near-perfect statistical correlations among Age, Experience, Salary, and Performance. While this demonstrates a highly structured and predictable organizational hierarchy, it highlights potential business risks regarding rigid tenure-based compensation and model multi-collinearity. ---

DATASET OVERVIEW

Dataset overview unavailable.

DATA QUALITY ANALYSIS

Data quality analysis unavailable.

STATISTICAL ANALYSIS

Statistical analysis unavailable.

BUSINESS INSIGHTS

Business insights unavailable.

RISK ASSESSMENT

Risk assessment unavailable.

CONCLUSION

Conclusion unavailable.

REPORT INFORMATION

Property	Value
Generated By	AI Data Analyst Agent
Developer	Pritiranjana Rout
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DATASET SUMMARY

Metric	Value
Rows	21
Columns	8
Health Score	92%
Duplicate Rows	1

DATASET HEALTH SCORE

Overall Health Score : 92%

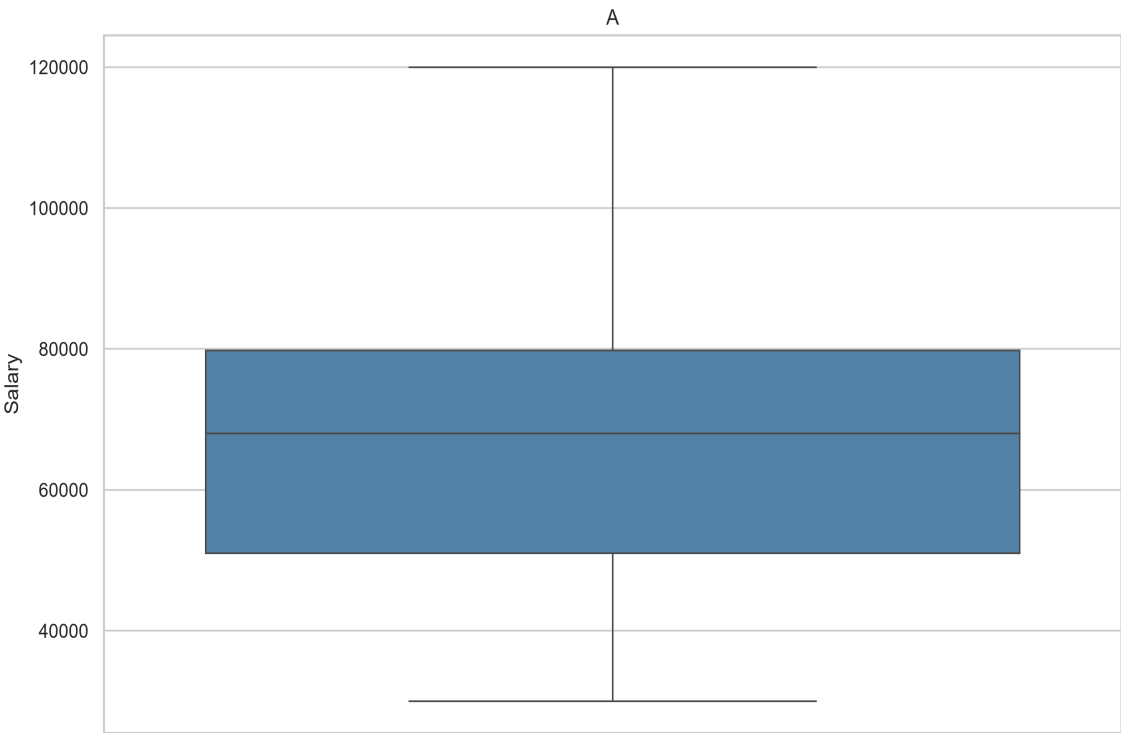
MISSING VALUES

Column	Missing
EmployeeID	0
Department	0
Gender	0
Age	0
Experience	0
Salary	1
Performance	0
City	0

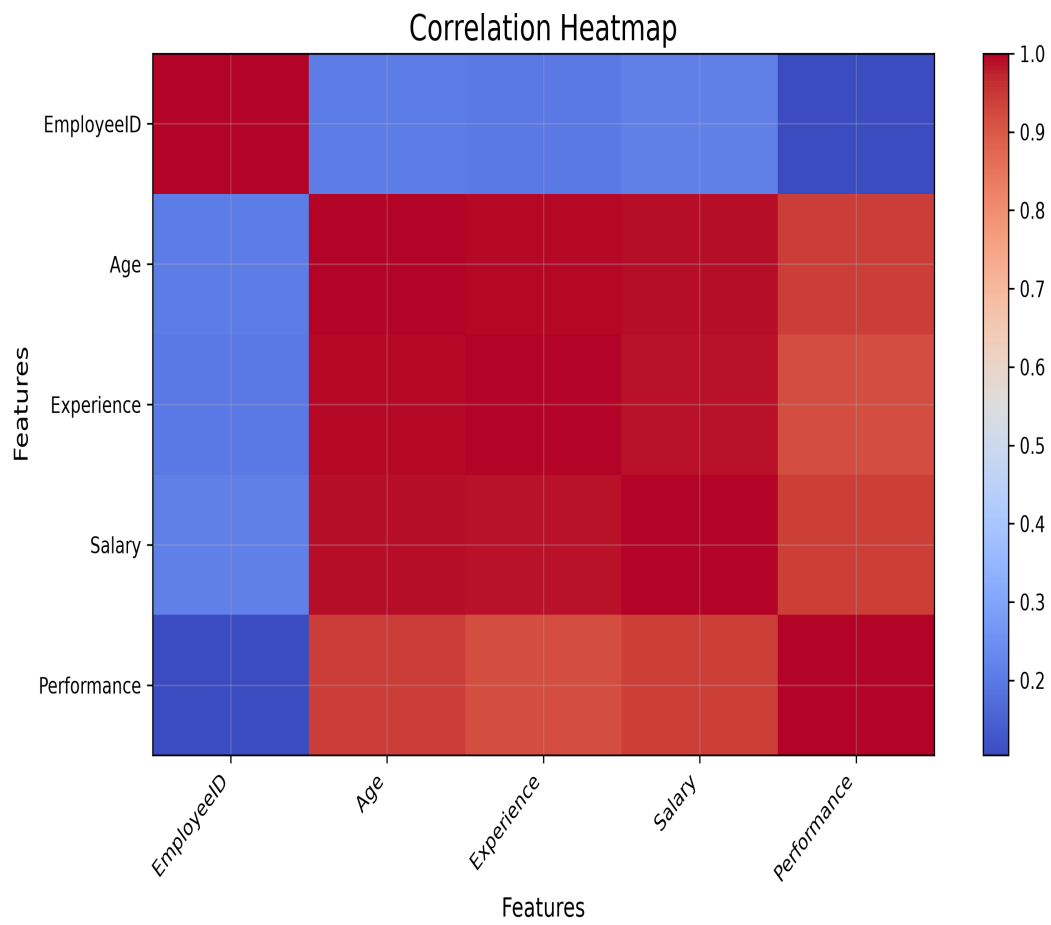
HISTOGRAM



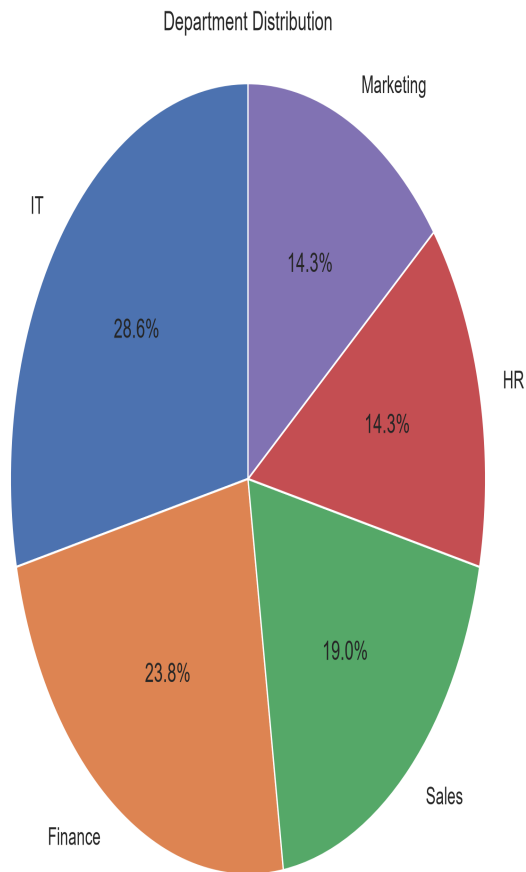
BOX PLOT



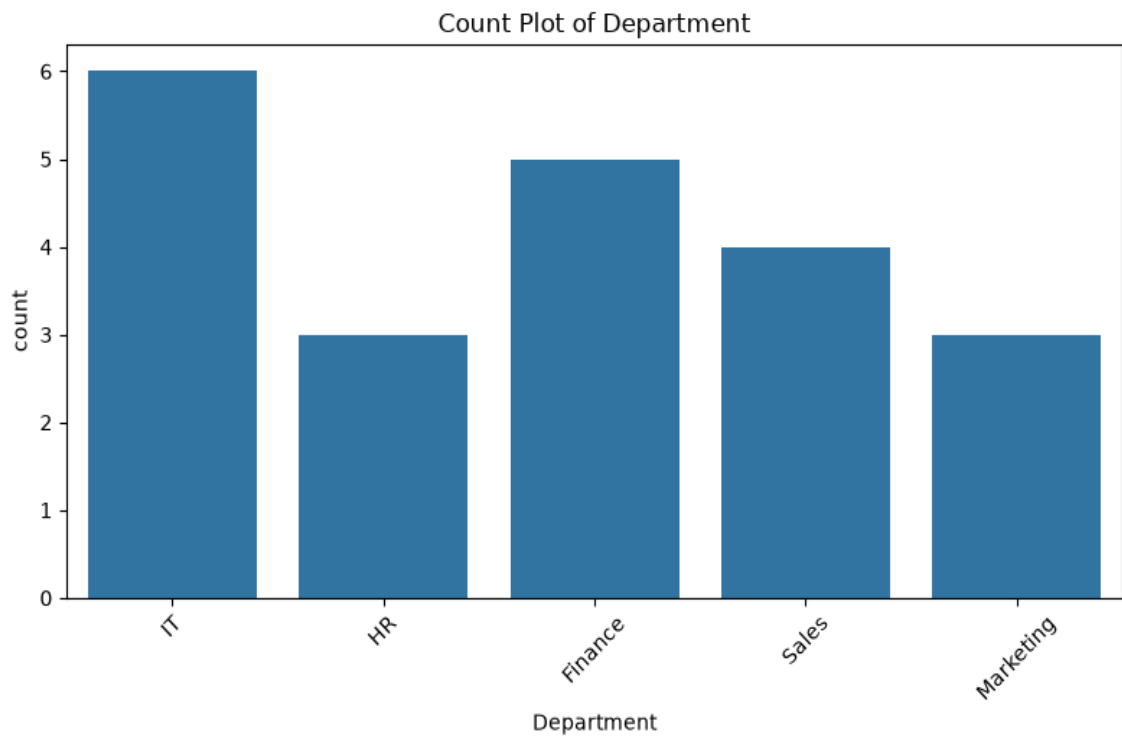
CORRELATION HEATMAP



PIE CHART



COUNT PLOT



OUTLIER ANALYSIS

Column	Outliers
EmployeeID	0
Age	0
Experience	1
Salary	0
Performance	0

STRONG CORRELATION ANALYSIS

Column 1	Column 2	Correlation
Age	Experience	0.99
Age	Salary	0.99
Age	Performance	0.94
Experience	Salary	0.99
Experience	Performance	0.92
Salary	Performance	0.94

Column 1	Column 2	Correlation
Age	Experience	0.99
Age	Salary	0.99
Age	Performance	0.94
Experience	Salary	0.99
Experience	Performance	0.92
Salary	Performance	0.94

AI RECOMMENDATIONS

- ■■ Dataset contains 1 missing values. Consider filling or removing them.
- ■■ Dataset contains 1 duplicate rows. Consider removing them.
- ■■ 1 potential outliers detected.
- ■ Strong correlations detected. A Correlation Heatmap is recommended.
- ■ Histogram, Box Plot and KDE Plot are recommended for numeric columns.
- ■ Pie Chart, Donut Chart and Count Plot are recommended for categorical columns.
- ■ Scatter Plot and Bubble Chart are recommended to analyze relationships.